MIL-P-63268A (AR)
4 JUNE 1981
SUPERSEDING
MIL-P-63268 (AR)
1 May 1978

MILITARY SPECIFICATION

PLUG, SHOCK, ATTENUATING

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacturing of Shock Attenuating Plugs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-1167	-Ammunition Data Card
MIL-STD-1169	-Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS (see 6.5)

ARRADCOM

9341742	- Plug, Shock Attenuating

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Plug. The plug shall comply with all requirements specified on the applicable drawings and in applicable specifications.

3.2.1 Plug, Shock, Attenuating. The plug shall comply with all requirements specified on Dwg. 9341742.

3.3 Tension. The plug shall sustain load of 8,000 pounds minimum without any sign of failure.

3.4 Ductility. The plug shall withstand deformation of the eye without cracking.

3.5 Fracture. The plug shall be free from porosity and shrinks or discontinuous structure.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, dust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Plug, Shock Attenuating	9341742	25

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63268A (AR)

PARAGRAPH	TITLE: Plug, Shock Attenuating		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Plug, Shock Attenuating Examination for Defects	25		3.2	4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Critical defects. See MIL-A-48078.
b. Major and Minor defects. See MIL-A-48078.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63268A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Plug, Shock Attenuating				9341742
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101W	Width of eye section, min.		0.40%	3.2	Gage
102	Length of thread section		0.40%	3.2	Gage
103	Dimension from bottom of thread section to bottom of flange		0.40%	3.2	Gage
104	Diameter of shank		0.40%	3.2	Gage
105	Dimension from bottom of gasket groove to face of flange, min.		0.40%	3.2	Gage
106	Diameter of flange, min.		0.40%	3.2	Gage
107	Diameter of gasket groove		0.40%	3.2	Gage
108	Depth of gasket groove, max.		0.40%	3.2	Gage
109	True position of gasket groove		0.40%	3.2	Gage
110	Perpendicularity of flange		0.40%	3.2	Gage
111	Thread pitch diameter, min.		0.40%	3.2	Gage
112	Thread pitch diameter, min.		0.40%	3.2	Gage
113	Tension test		4.4.3.1	3.3	4.5.1
114	Ductility test		4.4.3.1	3.4	4.5.2
115	Fracture test		4.4.3.1	3.5	4.5.3
116	Radius at base of shank, max.		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63268A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Plug, Shock Attenuating				9341742
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Thread pitch diameter, max.		0.65%	3.2	Gage
202	Width of gasket groove		0.65%	3.2	Gage
203	True position of flange		0.65%	3.2	Gage
204	Radius at top of shank, min.		0.65%	3.2	Gage
205	Outside radius of eye, min.		0.65%	3.2	Gage
206	Height of eye hole		0.65%	3.2	Gage
207	Width of eye hole		0.65%	3.2	Gage
208	Angle of groove inside diameter		0.65%	3.2	Gage
209	Identification notches missing		0.65%	3.2	Visual
210	Diameter of tool point, max.		0.65%	3.2	Gage
211	Radii or chamfers missing or incorrect		0.65%	3.2	Visual
212	Paint missing, incomplete or on undersurface		0.65%	3.2	Visual
213	Anti-seize compound missing or incomplete, as applicable		0.65%	3.2	Visual
214	Surface finish incorrect		0.65%	3.2	Visual
215	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.4.3 Testing. The tests (First Article and Quality Conformance) listed in the Item Detail Specification shall be performed on inspection lots as defined in 4.4.1 of this specification.

4.4.3.1 Mechanical properties tests. The sample size for these tests shall be 20 parts per lot. If any sample part fails to meet any requirement, the lot represented by the failing part shall be rejected.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test methods and procedures

4.5.1 Tension. The lifting plug shall be screwed into a threaded plate or other suitable fixture and a load of 8,000 pounds applied through the medium of a 3/4 inch diameter bolt or pin passing through the eye of the plug. The full load shall be transmitted through the screw threads of the plug.

4.5.2 Ductility. The plugs used in the tension test shall be cooled to minus 40 degrees Fahrenheit (F.) for a minimum of 30 minutes. The eye of each plug shall be flattened by either pressure or blows in any direction until the inner diameter of the eye is reduced by 0.050 inch while at minus 40 degrees F. Each plug shall be visually examined for cracks after flattening.

4.5.3 Fracture. The body of each sample malleable iron casting shall be broken through the thickest section for inspection. The plug shall be observed for discontinuous structure porosity or shrinks.

4.5.4 Dimensional control of cast or automatic tool parts. In place of the normal sampling associated with the classification of defects, and subject to approval of the Government procuring agency, the optional sampling plans in paragraphs 4.5.4.1 and 4.5.4.2 may be used for dimensional control of cast or automatic tool parts. Use of these optional plans is limited to those dimensions and features controlled by the tool, die, or mold cavity. For all specified tests, visual examinations, and examinations of dimensions or features not developed exclusively by the tool, die, or mold cavity, these optional plans are not applicable. In the event of in-process reject(s) during optional sampling, corrective action applicable to parts produced prior to the rejected, and subsequent to the last acceptable sample, shall require approval of the Government procuring agency.

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4.5.4.1 Cast parts. A sample of at least five parts (as cast) from each cavity shall be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided. If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above.

4.5.4.2 Automatic tool parts (see 6.9). A sample of at least 10 parts shall be dimensionally inspected to quality a new or adjusted setup for use in production. For multiple simultaneous operations (the same operation simultaneously performed on 2 or more parts), at least 5 parts per operation per setup shall be inspected. In addition, a random sample of 5 parts per setup or simultaneous operation shall be selected from the last portion of each hour's production, for dimensional inspection, as a control of the setup during production. If an hour's production is 250 parts or less per setup or simultaneous operation, the hourly sample may be reduced to 3 parts per setup or simultaneous operation. Automatic tools shall be considered for this reduced inspection only when equipped with devices to provide automatic shut-down or clear immediate warning if machine power input (electricity, air or oil pressure) falls below that minimum level necessary for normal machine operation. All parts in process when power input runs below minimum shall be inspected for dimensional requirements prior to acceptance. Such parts shall not be counted as qualification or control samples. If any defective parts are found during qualification of a setup, that setup shall not be used in production. If any defective parts are found when inspection is performed for control of a setup, the setup producing the defective part shall be removed from production. Further, that portion of production since the last setup control check shall be returned for corrective action. All setups removed from production because of some fault may, after adjustment or re-tooling as necessary, be returned to production providing they pass the qualification test above.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level C. Packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use in lifting and handling artillery projectiles.

6.2 Ordering data. see MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Data requirements. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Commmand (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.7 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

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6.8 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.9 Automatic tool. An automatic tool is any self-regulating piece or assemblage of production equipment which is setup or programmed to perform one or more operations on a part. The tool operator's functions shall not exceed part handling and input of a start signal. The tool setup shall provide positive, repetitive, three plane positioning of the work piece.

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